

Princy Reshma Ramaseshan

princyreshmar@gmail.com | (812) 822-7930 | LinkedIn | Github | Bloomington, IN

SUMMARY

Data Science graduate student with experience in **machine learning, NLP, and data analytics**, building ML pipelines, performing feature engineering and statistical analysis, and developing Python dashboards. Applied model training, evaluation, and validation through industry internships and research roles in **Data Analyst and Applied ML roles**.

EDUCATION

Master of Science in Data Science Indiana University, Bloomington	Aug 2024 – May 2026
<i>Coursework: Databases, Statistics, Applied ML, Data Visualization, Usable AI, Big Data, Deep Learning</i>	CGPA: 3.6/4
B.Tech in Computer Science SRM Institute of Science and Technology, Chennai	Aug 2020 – May 2024
<i>Coursework: Data Structures & Algorithms, Probability, Algorithms, Data Mining, DBMS</i>	CGPA: 3.88/4

WORK EXPERIENCE

Research Assistant Data Science and AI Lab, Kelley School of Business, IU	Jan 2026 – Present
<i>Python, Pandas, SQL, Hugging Face, GitHub APIs, Data Pipelines, Dashboards</i>	Bloomington, IN
– Execute and coordinate a large-scale Hugging Face–GitHub data collection pipeline using queue-based execution, API integration, and checkpointed workflows to identify machine learning models and analyze associated repositories for research analysis. – Contribute to interactive dashboard development using structured analytics, aggregated metrics, progress monitoring, and data visualization to track pipeline status, model coverage, and research insights in real-time production systems monitoring dashboards.	
Data Scientist My Youthspan Project, MyEdMaster	Aug 2025 – Dec 2025
<i>Python, Excel, Pandas, NumPy, Data Analysis, Research Databases, Literature Mining</i>	Leesburg, VA (Remote)
– Engineered a research database consolidating 9+ nutrition and performance studies by standardizing key metrics. – Analyzed literature papers using Python and Excel to quantify links between diet, biomarkers, and athletic performance outcomes.	
Research Assistant Faculty Assistance in Data Science, Indiana University	Jan 2025 – May 2025
<i>Python, NLP, Text Mining, NLTK, TF-IDF, D3.js, JavaScript, Data Visualization</i>	Bloomington, IN
– Performed large-scale NLP processing on 18.6M+ HathiTrust volumes, applying text normalization, tokenization, and corpus-level linguistic filtering to extract 111K+ keyword-in-context passages supporting scalable semantic analysis. – Designed and deployed interactive D3.js visualizations to facilitate pattern exploration, trend analysis, and insight discovery.	
Data Analyst Intern Soft Logic Systems Pvt. Ltd	Jun 2023 – Aug 2023
<i>Python, Pandas, NumPy, SQL, Data Preprocessing, ETL Pipelines, Data Cleaning, Dashboards</i>	Chennai, India
– Automated data preprocessing and transformation pipelines using Python and Pandas, reducing manual effort by 15+ hours. – Optimized SQL queries and designed interactive dashboards to enable data-driven business reporting and operational analytics.	

TECHNICAL SKILLS

Machine Learning & AI: Applied ML, NLP, Classification, Regression, Feature Engineering, Model Evaluation, model validation
Programming & Data Processing: Python, R, SQL, Pandas, NumPy, EDA, Data Cleaning, ETL Pipelines
ML / NLP Libraries: Scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face, NLTK, TF-IDF, word embeddings
Databases & Storage: MySQL, PostgreSQL, MongoDB, NoSQL, Structured and Semi-Structured Data, query optimization
Visualization & Analytics: D3.js, Plotly, Power BI, Tableau, Matplotlib, Seaborn, Dash, KPI tracking
Cloud & Tools: AWS, Google Cloud Platform, REST APIs, GitHub, Version Control, Jira, Teams, CI/CD pipelines

PROJECTS

Human Reference Atlas UI Usage Analysis (Client Project) Python, CloudFront, Plotly	Aug 2025 – Dec 2025
Analyzed 120K+ CloudFront logs to study user interactions across HRA tools by mapping raw HTTP requests into structured UI events. Built Plotly dashboards to track feature usage and inform data-driven interface optimization decisions insights.	
MindMender – Early Depression Detection from Reddit Posts Python, Dash, NLP	Jan 2025 – May 2025
Engineered and orchestrated an end-to-end NLP pipeline leveraging TF-IDF, Logistic Regression, and Sentence-BERT, achieving 75% accuracy and 0.76 macro F1 , while operationalizing feature extraction, and real-time Dash-based suicide risk alert.	
The Science of Happiness – Global Well-being Dashboard Python, Streamlit, Plotly, Pandas	Jan 2025 – Apr 2025
Built an interactive dashboard on 35K+ global well-being records using Pandas and Plotly for data processing and visualization. Leveraged choropleth maps and Sankey diagrams to analyze socio-economic trends, regional disparities, and correlation patterns.	

PUBLICATIONS

An Evaluation Methodology for News Headline Classification using Comprehensive Pattern Learning , IJFMR, Vol. 6, Issue 5, Sep–Oct 2024
Built a multi-label TF-IDF + SVM news classification system with K-Fold cross-validation, incorporating text normalization, n-gram feature extraction, and sparse vector representations to achieve **87% F1** on **10K+ real-world headlines**.

Online Auction Forecasting Precision: Real-time Bidding Insights and Price Predictions with Machine Learning , FMDDB Transactions, 1(2), 2023
Predicted auction sale prices on **10,681+ records** using XGBoost and regression models, engineering bid-history and temporal features, and achieving forecasting accuracy with **RMSE as low as 7.26** through bidder behavior modeling.